## Class 3

# The Recipe for Growth

*Part I — The Growth Miracle*

The American Economy — draft course materials (proposal under discussion)

# The point of today's class

*Growth is like a recipe: some gains come from more ingredients — workers, machines, schooling — but the most important long-run gains come from using those ingredients better: new ideas, better organization, and better institutions.*

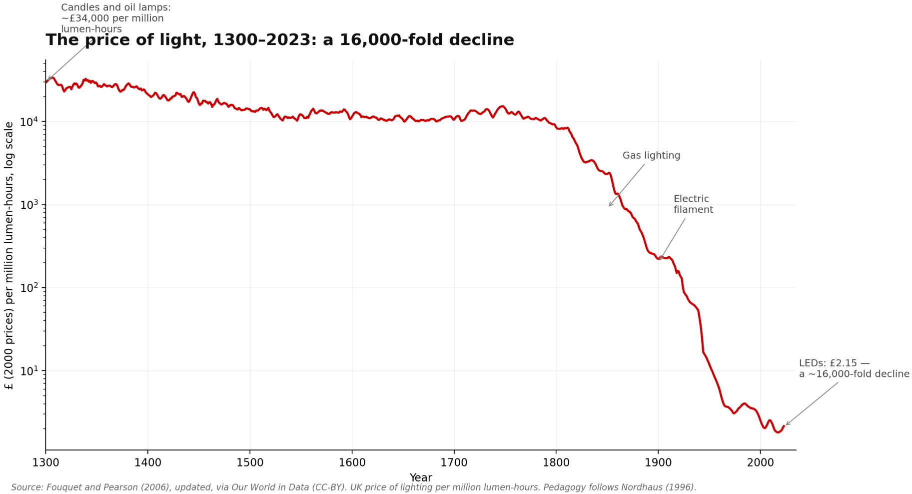
## Why it matters

- Class 2 showed that growth transformed the human condition. Today we ask *what the ingredients are*.
- What made America rich? Cotton fields, railroads, steel mills, the Model T line, Hoover Dam, land-grant universities, Bell Labs, immigrant workers, semiconductor fabs, AI data centers. . .
- Which of these is *the* answer? Today's claim: all of them — GDP is **produced**, not given.
- This class frames the whole course: every later class studies one ingredient in depth.

# Today

1. The recipe idea
2. The main fact: productivity rose enormously
3. More inputs: labor, capital, human capital, energy
4. Better recipes: ideas, innovation, organization
5. Growth accounting as a detective tool
6. The rest of the course as the ingredients list

# The price of light, 1300–2023: a 16,000-fold decline




*Source: Fouquet and Pearson (2006), updated, via Our World in Data (CC-BY). UK price of lighting per million lumen-hours, 2000 prices. Pedagogy follows Nordhaus (1996).*

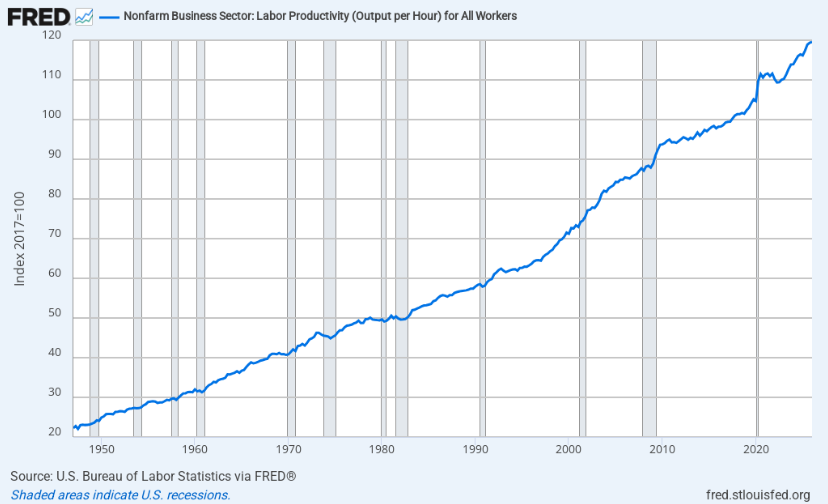
## The growth recipe, in words

- Flour + eggs + heat + instructions = cake. The economy's version, written once: $Y = A \times F(K, L, H, N, E, I)$

| Symbol | Plain English | U.S. examples |
|---|---|---|
| $Y$ | output / income | GDP, wages, consumption |
| $K$ | capital | railroads, highways, computers, data centers |
| $L$ | labor | population, immigration, hours worked |
| $H$ | human capital | literacy, schools, skills, health |
| $N$ | natural resources | land, coal, oil, rivers |
| $E$ | energy | steam, electricity, natural gas |
| $A$ | ideas / "how well the recipe works" | assembly line, germ theory, software, AI |
| $I$ | institutions | property rights, courts, patents, markets |

- Two ways to get more output: **more ingredients**, or a **better recipe** ($A$) — candles to LEDs is a better-recipe story.
- The ingredients **interact**: better institutions raise investment; education makes ideas usable; new ideas make capital more productive.

# U.S. labor productivity, 1947–present




*Source: Bureau of Labor Statistics, nonfarm business sector output per hour, via FRED (OPHNFB).*

## What does "output per hour" mean?

- Why can an hour of work today produce so much more than in 1947?
- Shovel vs. excavator: same worker, vastly more trench.
- Paper ledger vs. spreadsheet vs. AI: same accountant, vastly more accounts.
- "Productivity" sounds boring, but it is the main reason wages and living standards rise — the bridge from GDP growth to your paycheck.

# More inputs: America grew partly by having more of everything

- **Capital**: investment accumulates into factories, railroads, highways, machines.
- **Labor**: population growth, waves of immigration, women's rising participation.
- **Human capital**: mass schooling, land-grant universities, better health.
- **Energy**: steam, coal, electrification, oil.
- Nuance: more labor raises *total* GDP; living standards depend on output *per person* or *per hour*.
- Honest caveat: enslaved labor raised measured output for enslavers while violating self-ownership; the cotton gin raised productivity *and* expanded slavery. GDP is not welfare (Class 2) — the course returns to this in Class 10.

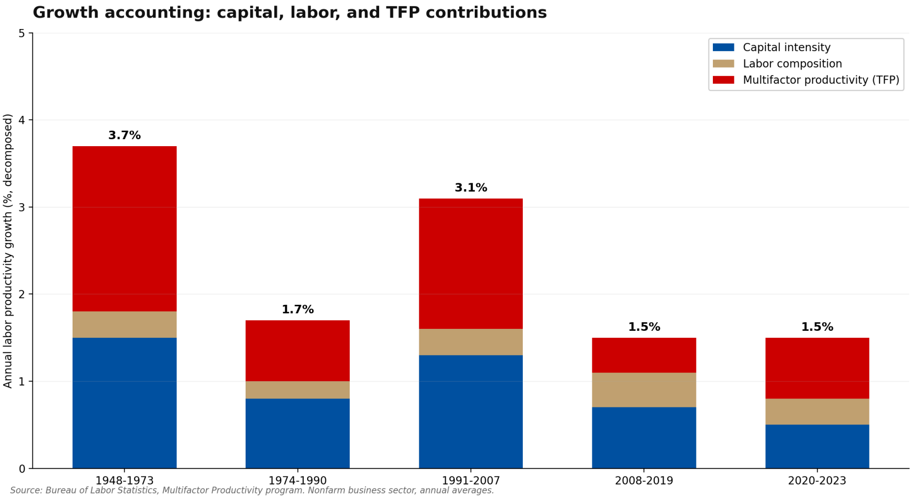
## Better recipes: ideas are different

- Romer: ideas are **nonrival** recipes — my using calculus does not use it up.
- Assembly line, germ theory, antibiotics, microchip, software, AI: each makes the *same* inputs produce more.
- Organization is technology too: Ford's line multiplied output per auto worker without adding workers.
- Are ideas getting harder to find? Research effort is way up; research productivity is falling (Bloom et al. 2020).
- The U.S. still accounts for a large share — roughly 60% — of world innovation.

## Three ideas hiding in the recipe

- **Allocation**: same workers, same machines, different placement — a subsistence farmer vs. an engineer designing bridges. Farm-to-factory migration, firm entry and exit, capital moving to better uses.
- **Scale**: a large integrated market made railroads, mass production, retail, and research pay off.
- **Complementarity**: electricity did little until factories were redesigned; computers needed software, management, and skills; AI needs chips, data centers, and electricity. New technology is slow to show up in productivity.

# Growth accounting: where does productivity come from?




*Source: Bureau of Labor Statistics, Multifactor Productivity program.*

## The residual: the "better recipe" part

- Growth accounting as a detective tool: output growth $-$ measured input growth $=$ the residual (Solow).
- The residual is the better-recipe part: technology — but also institutions, reallocation, scale, quality change, and mismeasurement.
- Measurement humility: do not call the residual "technology" too casually. Economists have not proved exactly where growth came from.

# The rest of the course is the ingredients list

- **Institutions**: liberty and the rules of the game (Class 4), the Constitution (Class 5), why nations diverge (Class 6).
- **Ideas**: innovation and the idea frontier.
- **Labor and human capital**: education, immigration, women's work.
- **Capital and organization**: firms, finance, infrastructure, government.
- If the recipe needs *rules* — who will invest, build, or lend, and under what incentives — the first ingredient to examine is the institutional one. Next class: liberty and property rights.

## Discussion

Which ingredients best explain America's success — and which create its hardest problems? Pick one ingredient and make the case.

## Read before next class

- Romer (1990), "Endogenous Technological Change" — read for the one idea: ideas are nonrival recipes.
- Bloom, Jones, Van Reenen, and Webb (2020), "Are Ideas Getting Harder to Find?" (skim the figures).
- Solow (1957), "Technical Change and the Aggregate Production Function" — optional; where growth accounting began.